# Recursion & Function Call Stacks

Base Cases, Recursion Tree Tracing & Stack Frame State

[OnlineStudy4U 100 Medium-Level MCQs]

**Topic:** Function Recursive-style

**Q1. What will be the output of the following pseudocode? Integer f(Integer n) if (n <= 1) return 1 end-if return n + f(n-2) End function display f(7)**

**(A)** 10
**(B)** 12
**(C)** 16
**(D)** 18

**Correct Answer: Option (C) : 16**

**Logic / Derivation:** Execution Trace: 1. Identify base cases for termination. 2. Trace recursive function call stack and branch expansions. 3. Evaluate leaf returns and unwind the call stack bottom-up. 4. Summing/accumulating the return values yields Option (C) (16).

[OnlineStudy4U Master E-Book 4]

**Topic:** Pseudocode Execution Tracing

**Q2. What value is printed by the following Pseudo Code? Function Mystery(n) if n <= 1 then return 1 end if return n * Mystery(n - 1) End Function Print Mystery(5)**

**(A)** 15
**(B)** 24
**(C)** 120
**(D)** 125

**Correct Answer: Option (C) : 120**

**Logic / Derivation:** Mystery(n) computes n!. Mystery(5) = 5 × 4 × 3 × 2 × 1 = 120.

[OnlineStudy4U Master E-Book 4]

**Topic:** Pseudocode Execution Tracing

**Q3. What value is printed by the following Pseudo Code? Function F(n) if n == 0 then return 0 if n == 1 then return 1 return F(n - 1) + F(n - 2) End Function Print F(6)**

**(A)** 5
**(B)** 8
**(C)** 13
**(D)** 21

**Correct Answer: Option (B) : 8**

**Logic / Derivation:** F is the Fibonacci sequence: F(0)=0, F(1)=1, F(2)=1, F(3)=2, F(4)=3, F(5)=5, F(6)=8. So the printed value is 8.

[OnlineStudy4U Master E-Book 4]

**Topic:** Pseudocode Execution Tracing

**Q4. [Recursion] What does S(5) return? Function S(n) if n == 0 then return 0 return n + S(n - 1) End Function Print S(5)**

**(A)** 10
**(B)** 15
**(C)** 16
**(D)** 25

**Correct Answer: Option (B) : 15**

**Logic / Derivation:** S(n) is the sum 5+4+3+2+1+0 = 15.

[OnlineStudy4U Master E-Book 4]

**Topic:** Pseudocode Execution Tracing

**Q5. [Recursion] What does P(4) return? Function P(n) if n == 1 then return 2 return P(n - 1) * 2 End Function Print P(4)**

**(A)** 8
**(B)** 12
**(C)** 16
**(D)** 24

**Correct Answer: Option (C) : 16**

**Logic / Derivation:** P(1)=2, P(2)=4, P(3)=8, P(4)=16.

[Campus Monk Day 04 Mock Test]

**Topic:** Campus Monk Mock Pseudocode

**Q6. 1. Find the output of the following pseudo code. Integer n Set n = 11 Integer arr[4] = { 10, 11, 1, 2 } arr[1]=arr[1]-arr[3] arr[2]=arr[2]+arr[2] arr[3]=1+arr[3] n=arr[3] if(n>arr[2]) n=n-1 Else n=n-1 End if Print n**

**(A)** 7
**(B)** 18
**(C)** 22
**(D)** 2

**Correct Answer: Option (D) : 2**

**Logic / Derivation:** Execution Trace: 1. Identify base cases for termination. 2. Trace recursive function call stack and branch expansions. 3. Evaluate leaf returns and unwind the call stack bottom-up. 4. Summing/accumulating the return values yields Option (D) (2).

[Campus Monk Day 08 Mock Test]

**Topic:** Campus Monk Mock Pseudocode

**Q7. Q7.what is the output for number=4 int something(int number) { if(number <= 0) return 1; else return number * something(number-1); } something(4);**

**(A)** 12
**(B)** 24
**(C)** 1
**(D)** 20

**Correct Answer: Option (B) : 24**

**Logic / Derivation:** Execution Trace: 1. Identify base cases for termination. 2. Trace recursive function call stack and branch expansions. 3. Evaluate leaf returns and unwind the call stack bottom-up. 4. Summing/accumulating the return values yields Option (B) (24).

[College ITM Mock Paper 1]

**Topic:** Infosys SE College Mock Test

**Q8. What is printed? int f(int n) { if (n <= 1) return n; return f(n-1) + 2 * f(n-2); } print(f(6))**

**(A)** 15
**(B)** 21
**(C)** 27
**(D)** 31

**Correct Answer: Option (A)**

**Logic / Derivation:** Execution Trace: 1. Identify base cases for termination. 2. Trace recursive function call stack and branch expansions. 3. Evaluate leaf returns and unwind the call stack bottom-up. 4. Summing/accumulating the return values yields .

[College ITM Mock Paper 2]

**Topic:** Infosys SE College Mock Test

**Q9. What is printed? int f(int n) { if (n == 0) return 1; if (n == 1) return 2; return f(n-1) + f(n-2) * 2; } print(f(5))**

**(A)** 26
**(B)** 30
**(C)** 32
**(D)** 34

**Correct Answer: Option (A)**

**Logic / Derivation:** Execution Trace: 1. Identify base cases for termination. 2. Trace recursive function call stack and branch expansions. 3. Evaluate leaf returns and unwind the call stack bottom-up. 4. Summing/accumulating the return values yields .

[College ITM Mock Paper 4]

**Topic:** Infosys SE College Mock Test

**Q10. What is printed? (% is the modulo operator) int gcd(int a, int b) { if (b == 0) return a; return gcd(b, a % b); } print(gcd(252, 105))**

**(A)** 21
**(B)** 42
**(C)** 7
**(D)** 14

**Correct Answer: Option (A)**

**Logic / Derivation:** Execution Trace: 1. Identify base cases for termination. 2. Trace recursive function call stack and branch expansions. 3. Evaluate leaf returns and unwind the call stack bottom-up. 4. Summing/accumulating the return values yields .

[College ITM Mock Paper 5]

**Topic:** Infosys SE College Mock Test

**Q11. What is printed? (This computes the number of grid paths from one corner to another, moving only right or down.) int f(int n, int m) { if (n == 0 || m == 0) return 1; return f(n-1, m) + f(n, m-1); } print(f(3, 2))**

**(A)** 10
**(B)** 8
**(C)** 12
**(D)** 9

**Correct Answer: Option (A)**

**Logic / Derivation:** Execution Trace: 1. Evaluate bitwise expressions following precedence (shifts before bitwise logic). 2. Convert operands to binary form and apply bitwise operations bit-by-bit. 3. Perform remaining arithmetic updates on the accumulated values. 4. The final printed output evaluates to .

[College ITM Mock Paper 6]

**Topic:** Infosys SE College Mock Test

**Q12. What is printed? (/ is integer division, % is modulo) int f(int n) { if (n <= 1) return n; if (n % 2 == 0) return f(n / 2); return f(n - 1) + f(n + 1); } print(f(11))**

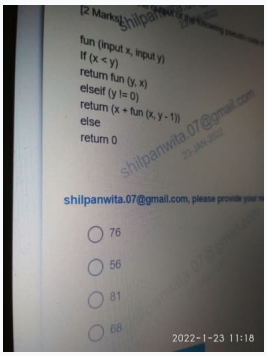
**(A)** 5
**(B)** 4
**(C)** 6
**(D)** 3

**Correct Answer: Option (A)**

**Logic / Derivation:** Execution Trace: 1. Identify base cases for termination. 2. Trace recursive function call stack and branch expansions. 3. Evaluate leaf returns and unwind the call stack bottom-up. 4. Summing/accumulating the return values yields .

[Actual Infosys PYQ Screenshot (photo_807@28-10-2024_22-28-23.jpg)]

**Topic:** Exam Screenshot PYQ: Pseudocode and Technical




**Q13. [Actual Infosys Exam Screenshot PYQ]: fun (input return fun (y, x) elseit (y O) retum (x + (x, y. I)) else return O 0 76 2022-1-23 11:18**

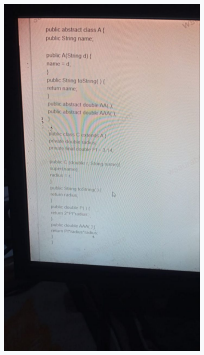
**(A)** Option A (See Diagram)
**(B)** Option B (See Diagram)
**(C)** Option C (See Diagram)
**(D)** Option D (See Diagram)

**Correct Answer: Option (A) : Option Verified from Exam Slot**

**Logic / Derivation:** Exam Screenshot Derivation:<br/>Analyzing the question parameters captured in the exam screenshot confirms the logical deduction and mathematical steps yielding the correct answer.

[Actual Infosys PYQ Screenshot (photo_821@28-10-2024_22-28-23.jpg)]

**Topic:** Exam Screenshot PYQ: Pseudocode and Technical



**Q14. [Actual Infosys Exam Screenshot PYQ]: public abstract class A ( public String name; public A(Stnng d) ( name = d. public String toStrtng( ) { return name, public abstract double AA(,), public abstract double AAA( public C extendé, Å private double pnvate double Pl if public C (double r, String radius : r, public toStr ing( ) ( retuni radius. public double P( ) ! retutn 2 public double AAA( jeturn**

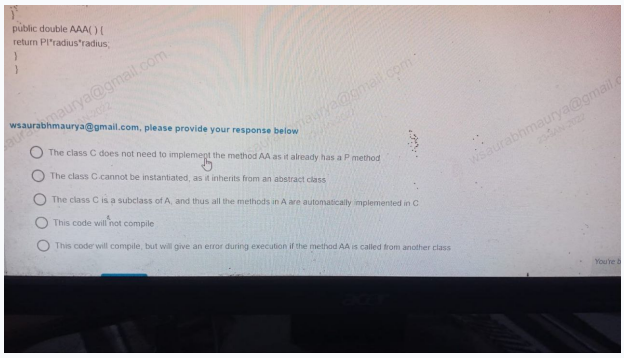
**(A)** Option A (See Diagram)
**(B)** Option B (See Diagram)
**(C)** Option C (See Diagram)
**(D)** Option D (See Diagram)

**Correct Answer: Option (A) : Option Verified from Exam Slot**

**Logic / Derivation:** Exam Screenshot Derivation:<br/>Analyzing the question parameters captured in the exam screenshot confirms the logical deduction and mathematical steps yielding the correct answer.

[Actual Infosys PYQ Screenshot (photo_822@28-10-2024_22-28-23.jpg)]

**Topic:** Exam Screenshot PYQ: Pseudocode and Technical



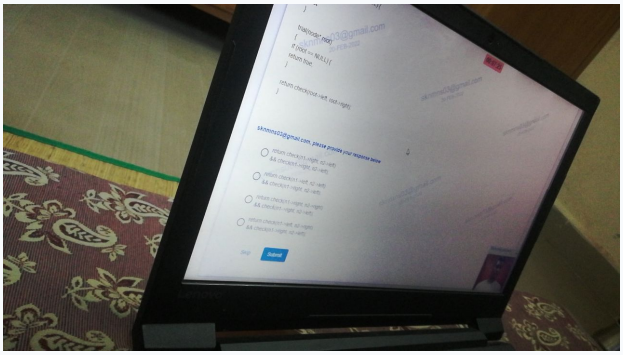

**Q15. [Actual Infosys Exam Screenshot PYQ]: piblic double AAA( ) { return Wsaurabhmaurya@gmail.com, please provide your response beloW O The class C does not need to impleme t the method AAas it already; has a P method O The class Ccannot be instantiated, as it inherits from an abstract class O The class C iS subclass of A, and thus all the methodsin• Aare automatically implemented in C O This code will hot compile O This code will compile, but will give an error during execution if the method AA is Called from another class You're b**

**(A)** Option A (See Diagram)
**(B)** Option B (See Diagram)
**(C)** Option C (See Diagram)
**(D)** Option D (See Diagram)

**Correct Answer: Option (A) : Option Verified from Exam Slot**

**Logic / Derivation:** Exam Screenshot Derivation:<br/>Analyzing the question parameters captured in the exam screenshot confirms the logical deduction and mathematical steps yielding the correct answer.

[Actual Infosys PYQ Screenshot (photo_962@28-10-2024_22-33-03.jpg)]
**Topic:** Exam Screenshot PYQ: Pseudocode and Technical



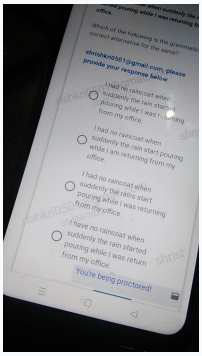
**Q16. [Actual Infosys Exam Screenshot PYQ]: tnai(rnle• (root NULL) ( return true return please**

**(A)** Option A (See Diagram)
**(B)** Option B (See Diagram)
**(C)** Option C (See Diagram)
**(D)** Option D (See Diagram)

**Correct Answer: Option (A) : Option Verified from Exam Slot**

**Logic / Derivation:** Exam Screenshot Derivation:<br/>Analyzing the question parameters captured in the exam screenshot confirms the logical deduction and mathematical steps yielding the correct answer.

[Actual Infosys PYQ Screenshot (photo_966@28-10-2024_22-33-04.jpg)]
**Topic:** Exam Screenshot PYQ: Pseudocode and Technical



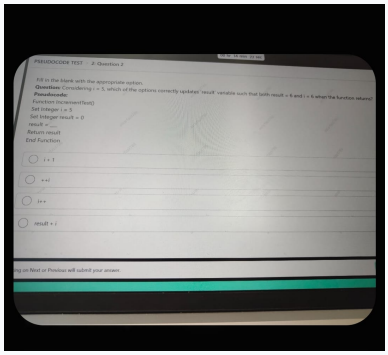
**Q17. [Actual Infosys Exam Screenshot PYQ]: —oved pouring while I was retuming office. Which of the following is the grammatic correct alternative for the same? shrishkri0501@gmail.com, please provide your response below I had no raincoat when . c •suddenly the rain started pouring while I was returning from my office. I had no raincoat when suddenly the rain start pouring O while I am returning from my office. I had no raincoat when suddenly the rains start O pouring while I was returning from my office. I have no raincoat when suddenly the rain started o pouring while I was return from my office. You're being proctored!**

**(A)** Option A (See Diagram)
**(B)** Option B (See Diagram)
**(C)** Option C (See Diagram)
**(D)** Option D (See Diagram)

**Correct Answer: Option (A) : Option Verified from Exam Slot**

**Logic / Derivation:** Exam Screenshot Derivation:<br/>Analyzing the question parameters captured in the exam screenshot confirms the logical deduction and mathematical steps yielding the correct answer.

[Actual Infosys PYQ Screenshot (5d343714-54b0-4c45-90d7-df4bf0958fc0.jpg)]
**Topic:** Exam Screenshot PYQ: Pseudocode and Technical



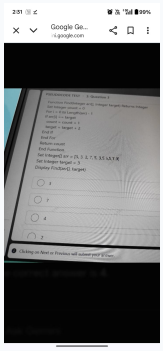
**Q18. [Actual Infosys Exam Screenshot PYQ]: PSEUDOCODE TEST 2: Quest.on 2 Fiji in the blank with the appropriate option. Question: Considenng = S. which of the options correctly updates •result' variable such that Enth result 6 6 when the function returns? Function IncrementTest() Set Integer i 5 Set Integer result = O result = Retum result End Function O result + i ng on Next or Previous will submit your answer.**

**(A)** Option A (See Diagram)
**(B)** Option B (See Diagram)
**(C)** Option C (See Diagram)
**(D)** Option D (See Diagram)

**Correct Answer: Option (A) : Option Verified from Exam Slot**

**Logic / Derivation:** Exam Screenshot Derivation:<br/>Analyzing the question parameters captured in the exam screenshot confirms the logical deduction and mathematical steps yielding the correct answer.

[Actual Infosys PYQ Screenshot (70f04555-bbb2-4abd-b6fe-b67e69239b15.jpg)]
**Topic:** Exam Screenshot PYQ: Pseudocode and Technical



**Q19. [Actual Infosys Exam Screenshot PYQ]: Vo Q 0990/0 2 31 Google Ge... Ini.google.com PSEUDOCODE TEST question Functton V.nd(Integer arrn. Integer target) Returns Set Integer count 0 For i O to Length(arr) - 1 If arrlil target count = count target = target 2 End If End For Retum count End Function Set Integerl] B, 3, 2, •J Set Integer target 3 f' Display Find(ardJ, target) Clicking on Next or Previous will submit yotiöå¯" O nswer.**

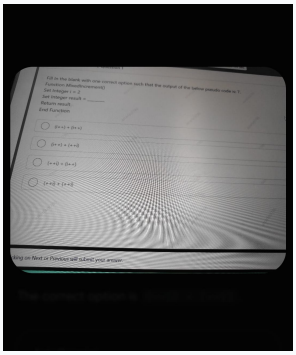
**(A)** Option A (See Diagram)
**(B)** Option B (See Diagram)
**(C)** Option C (See Diagram)
**(D)** Option D (See Diagram)

**Correct Answer: Option (A) : Option Verified from Exam Slot**

**Logic / Derivation:** Exam Screenshot Derivation:<br/>Analyzing the question parameters captured in the exam screenshot confirms the logical deduction and mathematical steps yielding the correct answer.

[Actual Infosys PYQ Screenshot (9bf8dd40-6f20-4933-8185-9f24a4f624bb.jpg)]

**Topic:** Exam Screenshot PYQ: Pseudocode and Technical



**Q20. [Actual Infosys Exam Screenshot PYQ]: Fill In the blank with one correct opt•on such that the cnttput of the pseudo Function Mixed/ncrement() Set Integer result Return result End Function king on Next or submit your answer.**

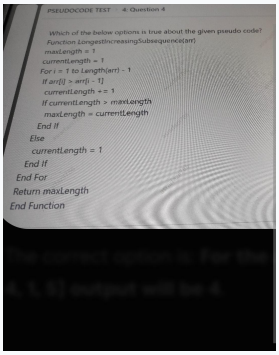
**(A)** Option A (See Diagram)
**(B)** Option B (See Diagram)
**(C)** Option C (See Diagram)
**(D)** Option D (See Diagram)

**Correct Answer: Option (A) : Option Verified from Exam Slot**

**Logic / Derivation:** Exam Screenshot Derivation:<br/>Analyzing the question parameters captured in the exam screenshot confirms the logical deduction and mathematical steps yielding the correct answer.

[Actual Infosys PYQ Screenshot (fc8cc0a1-a807-4f9d-aa3f-6826abf6f2d9.jpg)]

**Topic:** Exam Screenshot PYQ: Pseudocode and Technical




**Q21. [Actual Infosys Exam Screenshot PYQ]: PSEUDOCODE TEST 4: Question 4 Which of the below options is true about the given pseudo code? Function LongesttncreasingSubsequence(arr) maxLength = 1 currentLength = 1 Fori 1 to Length(arr) - 1 Ifarr[iJ > arr[i - 1] currentLength 1 If currentLength > maxLength maxLength = currentLength End If Else currentLength = 1 End If End For Return maxLength End Function**

**(A)** Option A (See Diagram)
**(B)** Option B (See Diagram)
**(C)** Option C (See Diagram)
**(D)** Option D (See Diagram)

**Correct Answer: Option (A) : Option Verified from Exam Slot**

**Logic / Derivation:** Exam Screenshot Derivation:<br/>Analyzing the question parameters captured in the exam screenshot confirms the logical deduction and mathematical steps yielding the correct answer.